
An Auditable Benchmark of Imbalance Handling for Multiclass Tabular Classification

Shaurya Saria
sariashaurya09@gmail.com

Abstract

We report a bounded comparison of fixed imbalance-handling conditions on 11 public multiclass datasets (9 numeric, 2 mixed) and three classifiers. The internally frozen protocol uses fold-local preprocessing, stratified five-fold evaluation under three seeds, complete dataset-level cells, and macro-F1 as the sole primary metric; it was not publicly preregistered. Of 5,445 enumerated records, 1,005 were unsupported combinations; of 4,440 applicable attempts, 4,191 were valid and 249 failed in the sampler. Four of 32 planned asymptotic Friedman tests were significant, all in the same numeric XGBoost block comprising the exactly balanced datasets CNAE-9, Iris, and Seeds. Seven of nine conditions tied exactly with raw on this block, while SMOTEENN and SMOTETomek occupied opposite rank extremes. A post hoc permutation sensitivity gave $p = 0.00012$, but with three blocks exact sign-flip tests have minimum two-sided $p = 0.25$. A separate post hoc analysis excluding ADASYN used eight conditions and nine numeric datasets; permutation p -values were 0.0018, 0.0476, and 0.0144 for logistic regression, random forest, and XGBoost, respectively, while none of the seven raw-baseline follow-ups per classifier survived Holm correction. The results show why applicability, failures, complete-cell coverage, and statistical resolution must accompany performance summaries.

1 Introduction

Class imbalance is a recurring source of difficulty in multiclass classification. A model can obtain a strong aggregate score while failing systematically on a rare class, and a resampling method that helps one classifier or feature representation can be neutral or harmful for another. This is especially important for tabular data, where the interaction between preprocessing, categorical representation, model inductive bias, and the sampling operator is part of the empirical question rather than a disposable implementation detail.

The literature contains mature families of imbalance interventions, including class weighting, random over- and under-sampling, synthetic minority generation, and combinations of over-sampling with cleaning or boundary-based methods (He & Garcia, 2009; Branco et al., 2016). Recent benchmark studies already cover multiclass imbalance, fold-local evaluation, and reproducibility at larger scale, including CLIMB and TILBench (Liu et al., 2025; Liu & Luo, 2026). We therefore make no novelty claim for multiclass scope, leakage-aware preprocessing, or reproducible benchmarking in general. Instead, we study a narrower fixed comparison and make applicability, failures, complete-cell coverage, and dataset-block inference explicit.

We ask how locked conditions compare with raw baselines; whether results differ by feature family and classifier; and what coverage, failure, runtime, and post-sampling row-count costs accompany the metric changes. Scope, conditions, metric definitions, and complete-case rules were frozen internally before the full run and analysis. There was no public registration or independent timestamped preregistration.

Our contributions are threefold:

- A bounded comparison of eleven public datasets, three classifiers, and feature-type-appropriate conditions, without a claim that this scope is unique.

- An explicit accounting of the full condition matrix, separating unsupported combinations from applicable evaluations and sampler failures.
- Dataset-level paired effects, limited-sample sensitivities, efficiency summaries, and an auditable source and artifact trail.

The resulting evidence supports a narrow conclusion: imbalance handling is conditional on the dataset, classifier, feature family, and metric, and the cost of a method is part of its practical evaluation. The paper does not claim a universally superior sampler.

2 Related Work

Imbalance interventions. SMOTE generates synthetic minority examples by interpolating between minority neighbours (Chawla et al., 2002); ADASYN adapts the amount of synthesis to local learning difficulty (He et al., 2008); and Borderline-SMOTE concentrates synthesis near difficult class boundaries (Han et al., 2005). Random over- and under-sampling provide simpler distributional changes. SMOTEENN and SMOTETomek combine over-sampling with cleaning or Tomek-link removal, respectively, and are representative of combination approaches studied in the imbalance literature (Batista et al., 2004). Our benchmark treats these methods as locked comparators, not as tunable algorithmic contributions.

Models and metrics. The model set intentionally spans a linear classifier, a bagged tree ensemble, and gradient-boosted trees. Balanced Random Forest is included as an RF-specific comparator following the idea of drawing balanced bootstrap samples for trees (Chen et al., 2004). XGBoost is a widely used gradient-boosting implementation for tabular prediction (Chen & Guestrin, 2016). We use macro-F1 as the primary metric and report multiclass G-mean, Matthews correlation coefficient, balanced accuracy, and per-class recall as secondary outcomes. This combination reflects the need to evaluate both class-wise performance and agreement beyond overall accuracy (Sokolova & Lapalme, 2009; Gorodkin, 2004).

Benchmarking and reproducibility. CLIMB and TILBench establish that broad empirical benchmarking of imbalanced tabular learning and multiclass evaluation are already represented in the literature. TILBench explicitly fits preprocessing and resampling on training data. The published CLIMB benchmark covers 73 real-world datasets and 29 methods across data-level, cost-sensitive, and ensemble approaches; it standardizes numeric features, encodes categories by cardinality, uses stratified five-fold evaluation, and searches up to 100 Optuna trials for each of 23 tunable methods per dataset (Liu et al., 2025). TILBench v2 covers 57 tasks (34 binary and 23 multiclass) and more than 40 methods; XGBoost is the reference learner for its data-level methods. It uses stratified 80:20 splits, reserves part of the training data for validation, runs 20 trials per method and five seeds, and reports mean and standard deviation for macro-F1 and G-mean (Liu & Luo, 2026). These studies are broader in dataset and method coverage. This study instead documents a narrow, fixed multiclass comparison: three classifier families, feature-specific numeric and mixed pipelines, no post hoc tuning, and row-level separation of unsupported combinations from sampler failures. Its aggregation preserves complete dataset-level cells and applies inference only when the frozen three-block rule is met. The contribution is an evidence-accounting comparison for this particular scope. Sampling operators come from the open-source imbalanced-learn ecosystem (Lemaître et al., 2017).

3 Benchmark Design

3.1 Datasets and scope

The benchmark was scoped to independent, public, multiclass tabular datasets with at least three classes, documented source terms, and sufficient support for five-fold stratification under a pre-run feasibility and scope audit (internally called Stage 0). The candidate registry contained 25 OpenML/UCI records. Eligibility and feature-type decisions were made from provenance, target, leakage, duplicate, missingness, and support audits; preliminary scores were not used to retain or remove datasets. Eleven datasets were retained: nine numeric datasets and two mixed datasets (CMC and Dermatology). Balanced and near-balanced tasks were not

Table 1: Scope and protocol differences between recent imbalance benchmarks and this study. Figures are reported by the cited works; the protocols are not directly comparable.

Study	Scope and experimental protocol	Reporting focus and relation to this study
CLIMB (NeurIPS 2025) (Liu et al., 2025)	73 real-world datasets, including binary and multiclass tasks; 29 methods spanning resampling, cost-sensitive learning, and ensembles. Numeric features are standardized; binary categories are ordinal-encoded and higher-cardinality categories one-hot encoded. Stratified five-fold evaluation; up to 100 Optuna trials for 23 tunable methods per dataset.	Reports cross-fold average performance plus runtime and robustness analyses; a broad benchmark and software resource.
TILBench v2 (Liu & Luo, 2026)	57 tasks (34 binary, 23 multiclass); more than 40 methods, with XGBoost as the reference learner for data-level methods. Stratified 80:20 holdout, a validation subset of training data, 20 Optuna trials per method, and five seeds. Preprocessing and resampling are fitted on training data; no numeric-versus-mixed sampler matrix is specified.	Reports mean and standard deviation for macro-F1 and G-mean, with global and family-level analyses.
This study	11 multiclass datasets (9 numeric, 2 mixed); logistic regression, random forest, and XGBoost; 9 numeric or 7 mixed core conditions. Stratified five-fold evaluation under 3 fixed seeds, without hyperparameter search; preprocessing is fold-local; numeric hybrids use SMOTE and mixed hybrids use SMOTENC internally.	Separates unsupported combinations from applicable attempts and sampler failures; aggregates complete dataset cells and applies inference only with at least 3 blocks. This is a narrow evidence-accounting comparison, not a new sampling method or comprehensive benchmark.

Table 2: Retained datasets and registry-level moderators.

Dataset	n	C	d	$n_{\min}$	Type	IR
Balance Scale	625	3	4	49	Numeric	5.88
CNAE-9	1,080	9	856	120	Numeric	1.00
Glass	214	6	9	9	Numeric	8.44
Iris	150	3	4	50	Numeric	1.00
Optdigits	5,620	10	64	554	Numeric	1.03
Seeds	210	3	7	70	Numeric	1.00
Wine	178	3	13	48	Numeric	1.48
Wine Quality Red	1,599	6	11	10	Numeric	68.10
Yeast	1,484	10	8	5	Numeric	92.60
CMC	1,473	3	9	333	Mixed	1.89
Dermatology	366	6	34	20	Mixed	5.60

excluded by the scope rule: the exact-balanced CNAE-9, Iris, and Seeds datasets and, for example, Optdigits (IR 1.03) and Wine (IR 1.48) meet the multiclass and support criteria and were retained independently of observed performance. Multiple Features (Factors) was deferred because its corrected replay took 1,043.5 seconds against the 900-second pre-run feasibility budget. This is a computational scope decision, not a score-based exclusion.

Table 2 reports the retained registry fields used in the analysis. The imbalance ratio is the majority-class count divided by the minority-class count. The retained collection spans 150–5,620 rows, 3–10 classes, 4–856 raw predictors, and imbalance ratios from 1.0 to 92.6. Minimum class support ranges from 5 to 554, making the complete-cell and failure rules important for interpreting the result coverage.

Supplementary files `data/dataset_registry.csv` and `data/acquisition_manifest.csv` identify the retained datasets by source ID and URL and record retrieval metadata, licenses or terms, target semantics, class counts, and SHA-256 hashes. Raw and processed data are not redistributed; the manifests provide the provenance needed to reacquire them under their stated terms.

Table 3: Locked comparison matrix and applicability rules.

Family	Conditions	Applicability
Baseline/weighting	Raw; class-weighted with the balanced class-weight rule	All feature families; model-specific class weights
Simple resampling	Random over-sampling; random under-sampling	All feature families
Numeric synthesis	SMOTE; ADASYN; Borderline-SMOTE (borderline-1); SMOTEENN and SMOTETomek with internal SMOTE	Numeric datasets; three neighbours
Mixed synthesis	SMOTENC; SMOTEENN and SMOTETomek with internal SMOTENC	Mixed datasets; ordinal categories at the sampler interface
Balanced ensemble	Balanced Random Forest	Random Forest only; 200 trees; replacement

3.2 Evaluation protocol

Each dataset uses stratified five-fold cross-validation with shuffling and seeds 0, 1, and 2. A required record is identified by dataset, classifier, condition, seed, and fold, yielding 15 seed-fold records per cell. Rows with missing targets are removed before splitting. Predictor imputation, encoding, scaling, resampling, and training-fold weights are fitted within each training fold. The held-out fold is only transformed and predicted; it is never resampled or used to fit a preprocessing operation.

Numeric pipelines impute numeric values with the training median and scale after imputation. Mixed pipelines impute numeric values with the training median and categorical values with the training mode, ordinal-encode categorical predictors before SMOTENC, and one-hot encode after resampling with unknown categories ignored. Numeric SMOTEENN and SMOTETomek use SMOTE internally; their mixed-data counterparts use SMOTENC internally. The mixed cleaning stages receive ordinal-coded categories, so their neighbour distances can impose an artificial ordering on category codes. Semantic categorical overrides are recorded separately from the registry so that numeric source codes are not treated as continuous by accident. This implementation detail limits how literally the mixed-hybrid results should be interpreted.

3.3 Models and conditions

The classifiers are logistic regression, random forest, and XGBoost. Logistic regression uses a maximum of 1,000 iterations. Random forest uses 200 trees and one worker. XGBoost uses 200 estimators, the histogram tree method, multiclass log loss for evaluation, and one worker. The locked conditions are summarized in Table 3; sampler parameters are deliberately simple and fixed rather than tuned after seeing the full-run scores.

The numeric core contains nine conditions for each classifier: raw, class-weighted, random over-sampling, random under-sampling, SMOTE, ADASYN, Borderline-SMOTE, SMOTEENN, and SMOTETomek. The mixed core contains seven conditions, replacing the ordinary synthetic methods with SMOTENC, SMOTEENN, and SMOTETomek. Balanced Random Forest is analysed only within the RF classifier and is kept separate from the general classifier comparisons.

3.4 Metrics and aggregation

Macro-F1 is the primary outcome, computed over all declared target classes with zero division set to zero. Secondary outcomes are multiclass G-mean, MCC, balanced accuracy, and per-class recall. If R_c denotes recall for class c , then

$$G = \left(\prod_{c=1}^C R_c \right)^{1/C}. \quad (1)$$

All encoded classes are included in Equation 1; a zero class recall makes G-mean exactly zero. MCC is the multiclass Matthews coefficient, balanced accuracy is the unweighted mean recall, and per-class recall uses the same declared labels and zero behavior.

Metric values are first averaged over the 15 seed-fold records within a complete dataset–classifier–condition cell. A cell is complete only if all 15 records are valid. The statistical block is the dataset, not the fold or seed. Planned Friedman tests and their Nemenyi follow-ups use only datasets with a complete cell for every condition in the family and selected classifier. The implementation formed each raw-versus-alternative comparison from complete cells for raw and that alternative only, so the dataset count can vary by comparison. For numeric XGBoost, most raw-baseline comparisons use nine dataset blocks, while ADASYN uses three. This pair-specific rule departs from the frozen protocol’s family-wide complete-case rule. We therefore report raw-baseline effect estimates, intervals, and tests as exploratory, not as planned confirmatory evidence. A Wilcoxon test was computed only when at least three paired dataset blocks were complete; comparisons with fewer blocks are descriptive.

3.5 Statistics and efficiency

For every non-baseline condition, the paired difference is the condition mean minus the raw-baseline mean for the same dataset, classifier, and metric. We report mean and median paired differences, win/tie/loss counts, paired rank-biserial correlation, and a 95% percentile bootstrap interval with 10,000 resamples and NumPy generator seed 0. Win/tie/loss uses the locked 10^{-12} tolerance, whereas Wilcoxon removes only exact-zero differences; a round-off-sized nonzero delta can therefore count as a descriptive tie while remaining in a test. Supplementary `wilcoxon_effects.csv` reports the effective nonzero count and minimum attainable p when the actual signed-rank calibration is exact. Bootstrap intervals are unadjusted, use dataset blocks, and are separate from the Holm-adjusted Wilcoxon procedure; they are descriptive with fewer than three blocks.

The frozen analysis plan uses two-sided tests with $\alpha = 0.05$. Friedman tests (Friedman, 1937) compare all conditions on family-wide complete blocks when at least three dataset blocks and three conditions are available. Significant Friedman tests are followed by the planned all-pairs Nemenyi procedure (Nemenyi, 1963) on the same blocks. The implementation also computed two-sided paired Wilcoxon signed-rank tests (Wilcoxon, 1945) for raw versus each alternative using pair-specific complete blocks, with Holm correction within each family, classifier, and metric. Because this pairwise rule departs from the frozen family-wide rule, these Wilcoxon and Holm results are exploratory. Eligible all-zero comparisons receive $p = 1$ and remain in their Holm family; rows with fewer than three complete pairs receive no test or adjusted p -value. Macro-F1 is the sole primary outcome; G-mean, MCC, and balanced accuracy are supporting secondary outcomes. Post hoc sensitivities include seeded within-block rank permutations and exact paired signed-rank sign flips, with their exchangeability assumptions stated alongside the results.

Runtime, peak resident memory, and training rows before and after sampling are descriptive efficiency outcomes. Imbalance ratio, class count, raw and encoded dimensionality, and feature type are exploratory moderators. With eleven datasets, moderator patterns are descriptive and cannot establish a selection rule.

3.6 Reproducibility boundary

The source package freezes the configuration, registry, acquisition manifest, environment record, scope lock, raw-run manifest, analysis manifest, and local analysis replay and validation report (internally called Gate E). The generated raw and analysis results remain ignored by Git but are hashed in the local release payload. The executable scripts and tests preserve the fold boundaries, validity accounting, aggregation, statistical rules, and figure generation. The paper is a draft for review; it does not imply external submission or venue acceptance.

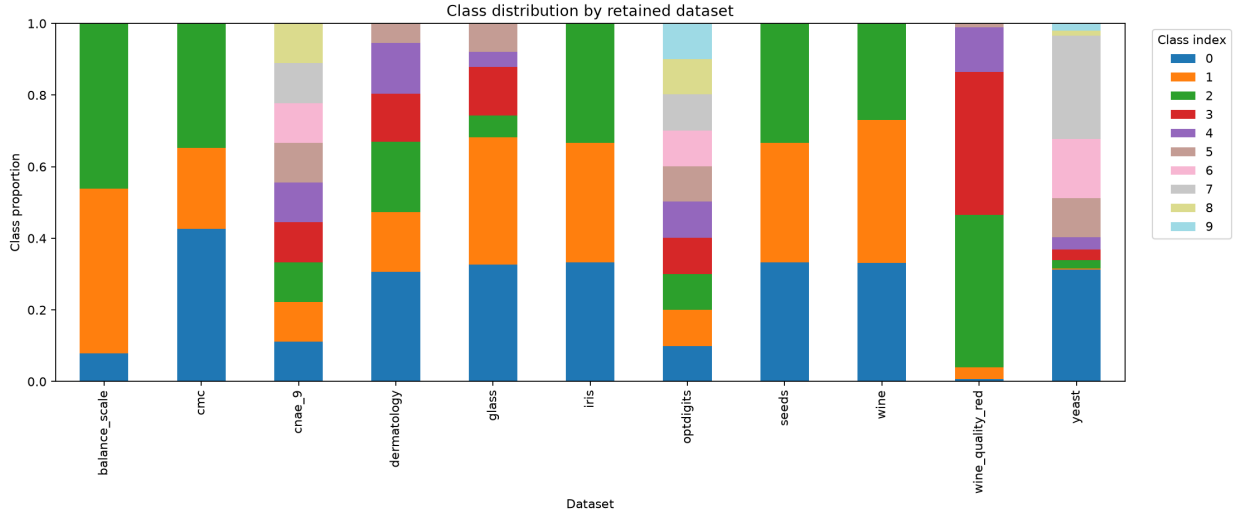


Figure 1: Class distributions for the 11 retained datasets; descriptive evidence for the moderator range.

4 Results

4.1 Coverage and failures

The full run enumerated 5,445 records. The applicability audit classifies 1,005 as unsupported combinations, leaving 4,440 applicable attempted evaluations: 4,191 valid records and 249 sampler-specific failures, with no other applicable failures. The sampler-specific failure rate is 5.61% of applicable attempts. The analysis contains 278 complete dataset–classifier–condition cells. Complete-cell counts were 75, 84, and 75 for logistic regression, random forest, and XGBoost in the numeric family, respectively; the corresponding mixed-family counts were 14, 16, and 14. The additional RF counts reflect the Balanced Random Forest comparator. These counts are not interchangeable with the number of folds: every complete cell contains 15 seed-fold records.

The 1,005 unsupported records comprise 405 numeric SMOTENC rows, 330 Balanced Random Forest rows outside the RF classifier, and 270 ordinary SMOTE, ADASYN, or Borderline-SMOTE rows on mixed data. The 249 sampler-specific failures are all numeric ADASYN: 207 reported that the requested ratio generated no samples and 42 reported no majority neighbours. These are distinct from unsupported combinations and remain visible in the raw failure log and failure-accounting table; neither category is scored as a model outcome.

4.2 Descriptive performance

Figure 2 groups mean macro-F1 by feature type and classifier (rows) and condition (columns). Each tile averages only complete dataset cells and has its own dataset coverage; a white blank marks no complete cell. The 66 tile counts are supplied in the supplementary coverage table. ADASYN has complete cells on CNAE-9, Iris, and Seeds for each classifier, and produced no additional training samples in any of the 135 valid seed-fold records across those cells. For the other six numeric datasets, 249 of 270 ADASYN records failed, leaving 21 valid records distributed across incomplete cells; those records remain in the audit but are not used as complete-cell outcomes. The complete-cell means are 0.9465 for logistic regression, 0.9332 for random forest, and 0.9242 for XGBoost. Each classifier therefore has only three complete ADASYN dataset blocks, with no paired macro-F1 differences from raw. These high means describe a narrow, balanced-data subset and do not establish general ADASYN superiority.

Among the numeric-core raw-baseline summaries with at least three complete pairs, the largest mean paired macro-F1 deltas were +0.0306 for logistic regression with random over-sampling, +0.0074 for random forest with SMOTETomek, and +0.0076 for XGBoost with class weighting. The table shows these three illustrative

Table 4: Largest mean paired macro-F1 deltas from raw among numeric-core summaries with at least three complete raw/alternative pairs, shown once per classifier. The 95% bootstrap intervals and Holm-adjusted p -values are exploratory because the pair-specific complete-case rule departs from the frozen family-wide rule.

Family	Classifier	Condition	n	Mean Δ	95% range	W/T/L	Holm p
Numeric	LR	Random over	9	+0.0306	[-0.0134, 0.0930]	3/3/3	1.000
Numeric	RF	SMOTETomek	9	+0.0074	[0.0016, 0.0148]	7/0/2	0.313
Numeric	XGBoost	Class weighted	9	+0.0076	[0.0015, 0.0150]	6/3/0	0.219

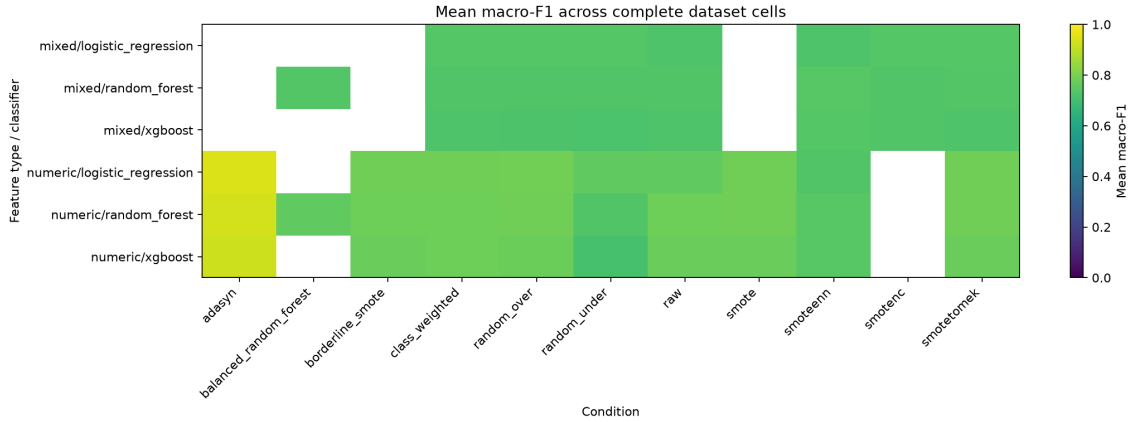


Figure 2: Mean macro-F1 across complete dataset cells, grouped by feature type and classifier (rows) and condition (columns). White blanks indicate no complete cell; counts are supplied in the supplementary coverage table. This display is descriptive; planned Friedman/Nemenyi tests use family-wide complete blocks.

rows, selected by the largest mean delta within each classifier; ties follow this locked order: class weighting, random over-sampling, random under-sampling, SMOTE, ADASYN, Borderline-SMOTE, SMOTEENN, then SMOTETomek. This is a descriptive display rule, not a method-selection procedure. The mixed-family largest means were +0.0127 for logistic regression with class weighting, +0.0149 for random forest with SMOTEENN, and +0.0113 for XGBoost with SMOTEENN, each based on only two dataset blocks. Those estimates remain descriptive; their bootstrap ranges are not calibrated intervals and they receive no inferential p -values.

4.3 Planned inference and exploratory results

There were 32 Friedman rows, four of which were significant. All four came from the same numeric-core XGBoost block on CNAE-9, Iris, and Seeds, each with IR 1.00 and nine conditions. Macro-F1, G-mean, MCC, and balanced accuracy each had statistic 24.0 and asymptotic $p = 0.002292$. These correlated outcomes on the same three datasets are not four independent confirmations; macro-F1 was the sole primary metric. In each metric, seven of the nine conditions (raw plus class weighting, random over- and under-sampling, SMOTE, ADASYN, and Borderline-SMOTE) tied exactly with raw on every block. SMOTEENN and SMOTETomek had the opposite mean-rank extremes, 9.0 and 1.0, respectively; their planned Nemenyi comparison was the only $p < 0.05$ pair ($p = 0.010464$). On these balanced data, the result chiefly distinguishes the two cleaning pipelines; it is not evidence of improved minority-class rebalancing. The exact-tie pattern is specific to XGBoost: on the same balanced cells, random-forest class weighting and random undersampling had nonzero macro-F1 differences from raw.

The post hoc rank-permutation sensitivity for the original three-block XGBoost macro-F1 matrix used 200,000 seeded permutations and returned 23 exceedances, $p = 0.000120$ (Monte Carlo SE 0.000024). Exact two-sided sign flips enumerate only eight patterns per paired contrast; the minimum attainable p is $2/8 = 0.25$, so this test cannot reject at 0.05 even before Holm correction. Its non-rejection is limited resolution, not contrary evidence. No pair survived Holm correction (smallest adjusted $p = 1.0$). The omnibus permutation assumes condition-label exchangeability within each selected dataset under its null; the paired sign-flip procedure

assumes independent sign exchangeability of dataset-level differences. Both are post hoc sensitivities on three blocks, not replacements for the planned asymptotic tests. The remaining 28 Friedman rows were non-significant or skipped; mixed-family tests had only two complete blocks and were skipped under the three-block rule.

The retained raw-baseline table has 184 rows: 46 per metric, comprising 24 numeric-core, 18 mixed-core, two numeric Balanced-RF, and two mixed Balanced-RF comparisons. Of these, 104 rows have at least three pair-specific complete blocks and received Wilcoxon tests and Holm correction; 80 rows with one or two blocks are descriptive. Because the implementation used pair-specific complete cases instead of the frozen family-wide rule, all 104 tests and adjusted values are exploratory, not planned confirmatory evidence. Twelve eligible all-zero ADASYN comparisons receive $p = 1$ and remain in their eight-test Holm families. The supplementary table reports each test’s effective nonzero count and, where exact calibration applies, its minimum attainable p . One adjusted value was below 0.05: for numeric logistic regression and MCC, SMOTEENN had nine nonzero differences (nine losses), mean delta -0.0667 , median delta -0.0265 , rank-biserial correlation -1.0 , exact two-sided $p = 0.003906$ (minimum attainable $p = 0.003906$), and Holm-adjusted $p = 0.03125$. Its unadjusted 95% paired-block bootstrap interval was $[-0.1105, -0.0283]$; bootstrap intervals and Holm-adjusted tests are distinct procedures. This exploratory result is not a planned confirmatory finding or sampler recommendation. In mixed XGBoost balanced accuracy for SMOTENC, the two pairwise deltas are $+0.007018053$ (CMC) and -2.22×10^{-16} (Dermatology): the locked 10^{-12} W/T/L tolerance counts one win and one tie, while exact-zero Wilcoxon preprocessing would retain both nonzero values. With only two complete pairs, this row is descriptive and no Wilcoxon test is run; the frozen tolerance and zero rules are unchanged.

A second post hoc sensitivity excludes ADASYN from the numeric condition set. All three classifiers then have the same nine complete numeric datasets and eight conditions. The asymptotic Friedman statistics and seeded permutation p -values were 20.511 and 0.001845 for logistic regression, 13.749 and 0.047595 for random forest, and 16.466 and 0.014400 for XGBoost; the corresponding asymptotic p -values were 0.004566, 0.055831, and 0.021183. Classifier-specific permutation sensitivities are not adjusted across classifiers. Exact signed-rank sign flips for the seven raw-baseline follow-ups per classifier, with Holm correction within each seven-test family, produced no adjusted $p < 0.05$ (the smallest was 0.136719 for XGBoost/SMOTEENN). These post hoc results change both the condition set and dataset coverage; they do not isolate a causal explanation, and non-significance does not establish equivalence.

For descriptive context, the post hoc numeric stratum at IR 1.00 contains CNAE-9, Iris, and Seeds (3 datasets); the IR > 1.00 stratum contains Balance Scale, Glass, Optdigits, Wine, Wine Quality Red, and Yeast (6). The supplementary table reports each classifier-condition mean and median paired macro-F1 delta for these strata. No inference is drawn from the three-dataset balanced stratum, and the full range of imbalance in the other six datasets makes that group descriptive rather than a homogeneous regime.

The class-wise results are reported in the analysis package rather than collapsed into one cross-dataset class index, because class labels and class semantics differ across datasets. The per-class recall summaries provide descriptive evidence for which classes drive a macro metric while preserving the dataset-specific label space.

4.4 Efficiency costs

Sampling changed the number of training rows in the expected direction, but the magnitude depended strongly on the dataset. Across raw numeric cells, the average training set size before sampling was 992 rows. SMOTE increased this to an average of 1,548 rows, random over-sampling had the same average post-sampling size, SMOTETomek averaged 1,518 rows, and SMOTEENN averaged 1,247 rows; random under-sampling reduced it to 661 rows. These are descriptive averages over dataset-level efficiency rows, not a claim that every dataset follows the same pattern.

The average fit-and-predict time for numeric random forest was 0.474 seconds for the raw baseline and 0.625 seconds for SMOTETomek; Balanced Random Forest averaged 0.721 seconds. For numeric XGBoost, the raw baseline averaged 0.538 seconds, while SMOTETomek averaged 0.737 seconds. The numeric ADASYN mean for XGBoost was 0.860 seconds, again based on only three complete datasets. In the two mixed datasets, RF averaged 0.365 seconds for the raw baseline, 0.501 seconds for SMOTENC, and 0.538 seconds for Balanced

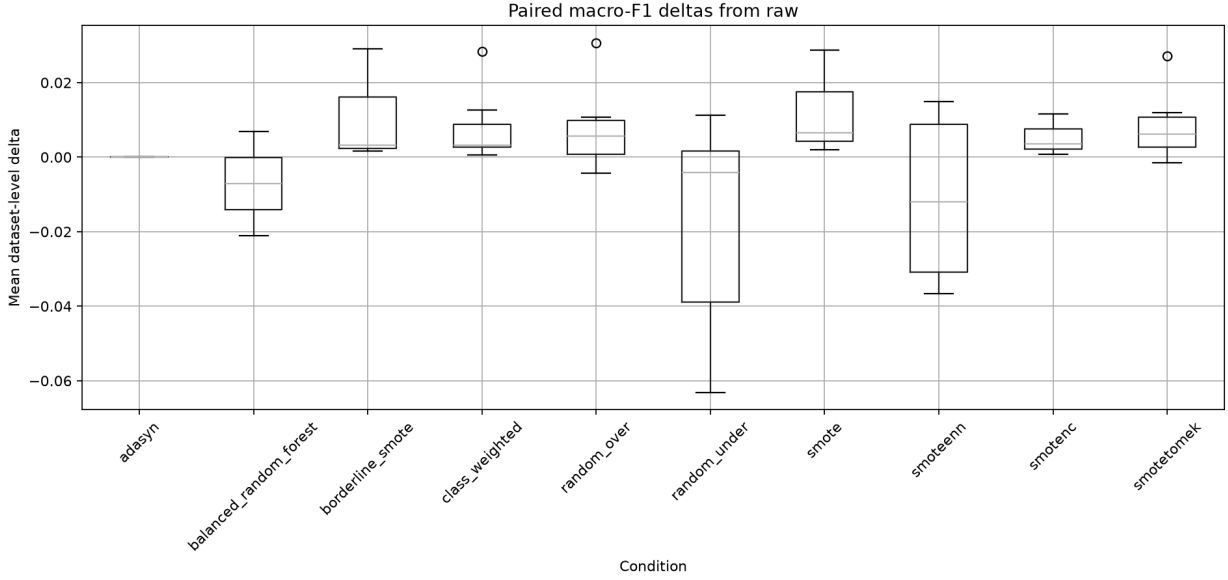


Figure 3: Distribution, by condition, of mean paired macro-F1 deltas from raw across classifier-by-analysis-family rows. Each observation is one row’s mean over its complete raw/alternative dataset pairs; it is not an individual dataset delta. Rows with one or two pairs are included as descriptive summaries. The center line marks the median, boxes span the first and third quartiles, and whiskers extend to the most extreme observations within 1.5 interquartile ranges; points beyond the whiskers are outliers.

Random Forest. These measurements support treating computational cost and applicability as first-class benchmark outcomes.

5 Discussion

The most interpretable planned separation occurs on three exactly balanced numeric datasets: SMOTEENN and SMOTETomek differ in rank while seven of nine XGBoost conditions reproduce raw macro-F1 exactly. The distinction concerns the behavior of two cleaning pipelines on balanced controls, not successful rebalancing of minority classes. Random-forest weighting and undersampling show that this exact-tie pattern does not hold for every classifier. The nine-dataset ADASYN-excluded sensitivity provides broader numeric coverage; its permutation omnibus results are small-sample sensitivities, and none of its exact raw-baseline follow-ups survives its within-classifier Holm correction. The asymptotic and permutation calibration differ for random forest, underscoring the limited resolution of this analysis. The separate original raw-baseline table remains exploratory because of its pair-specific complete-case departure; mixed-family summaries remain descriptive with only two blocks. Together, the results support reporting coverage and failure denominators alongside performance rather than treating incomplete-cell means as an unconditional ranking.

The benchmark also shows why feature type should be part of the protocol. Ordinary SMOTE-family conditions are marked unsupported for mixed datasets; mixed data uses SMOTENC before one-hot encoding. The hybrid cleaners receive ordinal-coded categorical columns, so nearest-neighbour distances in their cleaning stage can reflect arbitrary code order. The mixed family also has a different comparison matrix and only two retained dataset blocks. Pooling numeric and mixed conditions would violate applicability or conceal that the mixed inference block is too small for the planned tests.

The efficiency evidence reinforces this interpretation. Over-sampling usually increased the number of training rows, and more elaborate conditions generally increased fit-and-predict time relative to the raw baseline. The effect is not a universal penalty: random under-sampling reduced row counts and often reduced runtime, but its performance deltas varied by classifier and dataset. A practical choice therefore involves a task-specific trade-off among class-wise performance, failure risk, resource cost, and reproducibility.

5.1 Ethical considerations and broader impact

Two retained datasets come from sensitive health contexts. CMC is a subset of the 1987 National Indonesia Contraceptive Prevalence Survey, with demographic and socioeconomic predictors and contraceptive method as the outcome (Lim, 1999). Dermatology contains patient clinical and histopathological attributes; its UCI description reports that patient names and identification numbers were removed (Ilter & Guvenir, 1998). This study reuses existing public records for aggregate methods benchmarking; it does not recruit participants or evaluate a model for deployment. Public availability alone does not establish consent for every downstream use or an ethics-board determination for a particular secondary analysis, and we do not infer either from the dataset licenses. The benchmark does not assess clinical validity, privacy risk, or subgroup fairness. Its results should not guide contraceptive, clinical, or other high-stakes decisions; any such use requires separate review of data authority and consent, privacy, subgroup performance, and applicable domain and institutional oversight.

5.2 Limitations

First, eleven datasets do not establish a universal method ranking, especially for the two-dataset mixed family. The significant planned XGBoost block contains only three exactly balanced datasets, so it provides little resolution about performance under severe imbalance. The ADASYN-excluded sensitivity increases numeric coverage to nine datasets but is post hoc, changes the condition set, and does not establish equivalence or isolate a causal mechanism. Second, the candidate pool and retained scope are fixed by the project protocol and do not represent all tabular domains. Third, sampler and classifier hyperparameters are locked baseline settings rather than a full tuning study; tuning could change both performance and cost. Fourth, the complete-cell rule is conservative but reduces the inferential sample when a method fails or is inapplicable. Fifth, the mixed hybrid cleaner distance caveat limits interpretation of SMOTEENN and SMOTETomek on categorical data. Sixth, the exploratory imbalance strata are descriptive and should not be used as a selection rule. Stored-record validation is not an independent model-training replication, and independent human technical review remains outstanding.

5.3 Reproducibility and future work

The source and evidence package records the protocol, registry, acquisition provenance, executable benchmark and analysis scripts, focused tests, generated-figure definitions, and release validators. The analysis manifest records runtime package versions and SHA-256 hashes for its tables and figures; an analysis replay independently regenerates and compares those outputs. The supplement locates complete-cell coverage, class-wise recall, failure accounting, and the ADASYN-excluded sensitivity in named CSV files. Reprocessing the frozen result and failure records is an analysis-only reproduction; it does not rerun model training or constitute an independent replication. Future work could enlarge the retained dataset collection, implement categorical-only handling, add a separately timestamped protocol for tuning budgets, and test whether the observed conditionality persists under other model families. Such extensions would require a new scope or protocol amendment rather than being folded into this result set.

6 Conclusion

We presented an auditable, fixed-scope comparison of multiclass imbalance handling that links performance summaries to applicability, failures, dataset-level coverage, and computational cost. The planned XGBoost rank separation is concentrated in two cleaning methods on three exactly balanced datasets, while the post hoc nine-dataset sensitivity found no Holm-adjusted raw-baseline difference. These results support conditional

interpretation within the retained data and the declared statistical resolution, not a universal method ranking. The released analysis procedures reproduce the reported tables from frozen records; an independent full model-training replication has not been established.

7 Data and Code Availability

The anonymous source and evidence supplement contains the analysis code, frozen result and failure records, manifests, tables, figures, and reproduction instructions. `data/dataset_registry.csv` and `data/acquisition_manifest.csv` identify each retained dataset and its source terms. In `results/analysis/`, `macro_f1_figure_coverage.csv` gives complete-cell coverage; `class_means.csv`, `per_class_recall_summary.csv`, and `baseline_deltas_per_class.csv` contain class-wise summaries; and `failure_accounting.csv` plus `failure_reason_counts.csv` report coverage and failure categories. `adasyn_excluded_friedman.csv`, `adasyn_excluded_pairwise.csv`, and `adasyn_excluded_imbalance_descriptive.csv` report the post hoc sensitivity. The raw datasets are not redistributed. Running the analysis command regenerates the tables and figures from the supplied frozen result records; it does not rerun benchmark models or establish an independent training reproduction.

The Git remote URL for the benchmark source repository is <https://github.com/icecold009/multiclass-imbalance-benchmark>. The immutable source commit used to generate the frozen full-run records is <https://github.com/icecold009/multiclass-imbalance-benchmark/commit/18e3d1c1c82c2b0e1b911553c576de41b9a151bc>. That commit identifies the benchmark-run source; it does not contain the manuscript and corrected analysis revision reported here. The corrected manuscript and analysis revision are not yet in an immutable public release, and no DOI-backed archive is cited at this draft stage.

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